Supporting Information

Machine Learning Accelerated Design of Self-Assembled Monolayers for High-Performance Perovskite Solar Cells

Haifeng Li, Yue Zang,* Zhikang zhu, Chenyang Zhu, Weihong Liu, Wensheng Yan*

Institute of Carbon Neutrality and New Energy, School of Electronics and Information

Hangzhou Dianzi University

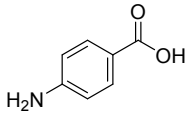
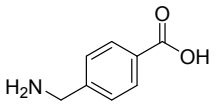
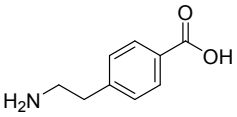
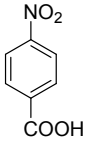
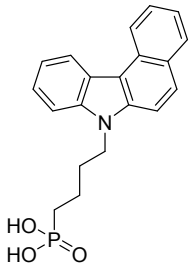
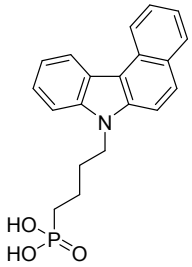
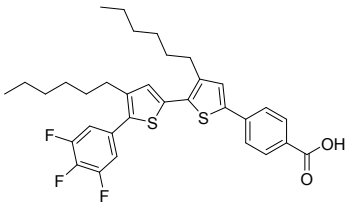
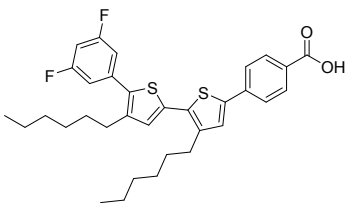
Hangzhou 310018, PR China

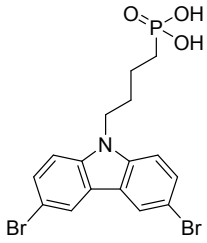
Corresponding authors:

zangyue@hdu.edu.cn (Y. Zang),

wensheng.yan@hdu.edu.cn (W. Yan)

Table S1 Nine low-PCE data points that were removed

SAM Name	SMILES	Molecular Structure	Experimental PCE（%）
AB	NC1=CC=C(C(O)=O)C=C1		7.6[1]
AM	NCC1=CC=C(C(O)=O)C=C1		5.3[1]
AE	NCCC1=CC=C(C(O)=O)C=C1		5.2[1]
NB	O=C(C1=CC=C([N+]([O-])=O)C=C1)O		4.5[1]
BCB-C4PA	OP(CCCCN1C2=C(C(C=CC=C3)=C3C=C2)C4=C1C=CC=C4)(O)=O		4.7[2]
BCB-C4PA	OP(CCCCN1C2=C(C(C=CC=C3)=C3C=C2)C4=C1C=CC=C4)(O)=O		9.7[2]
EA-50	FC1=C(F)C(F)=CC(C2=C(CCCCCC)C=C(C3=C(CCCCCC)C=C(C4=CC=C(C(O)=O)C=C4)S3)S2)=C1		9.09[3]
EA-56	FC1=CC(F)=CC(C2=C(CCCCCC)C=C(C3=C(CCCCCC)C=C(C		8.71[3]

	4=CC=C(C(O)=O)C= C4)S3)S2)=C1		
2BrCzPA	BrC(C=C1)=CC2=C1 N(CCCCP(O)(O)=O) C3=C2C=C(Br)C=C3		4.24[4]

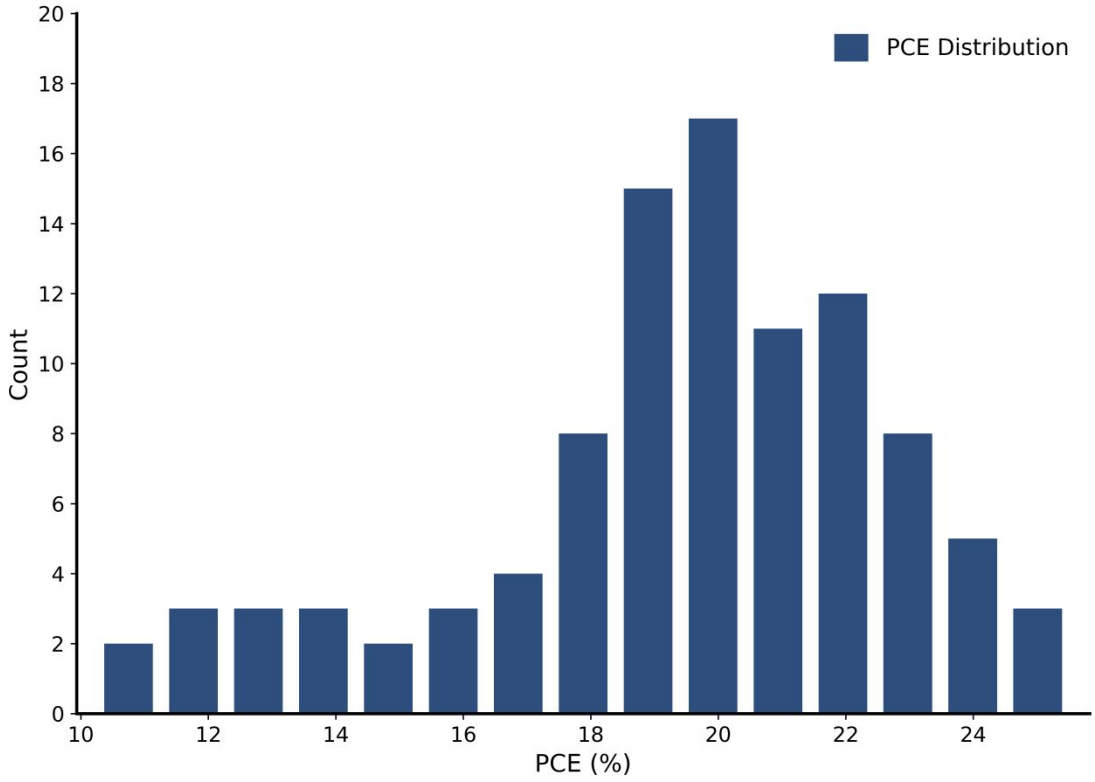
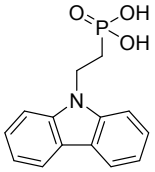
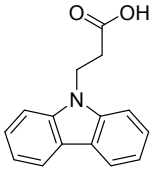
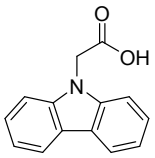
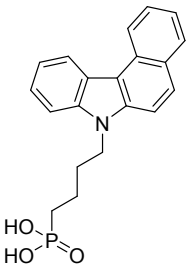
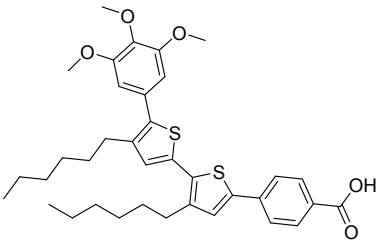
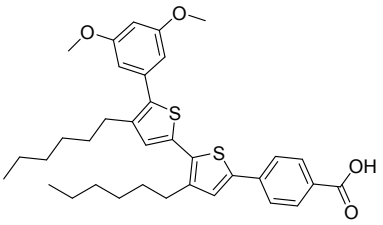
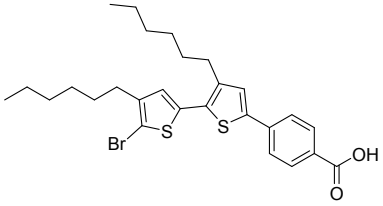
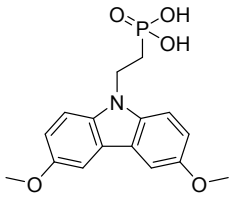
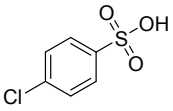
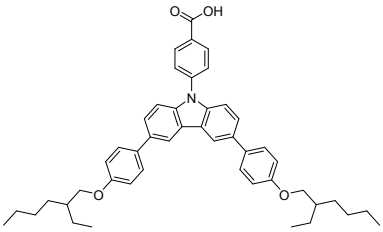
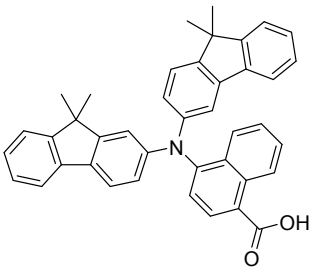
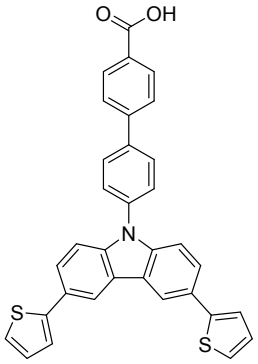
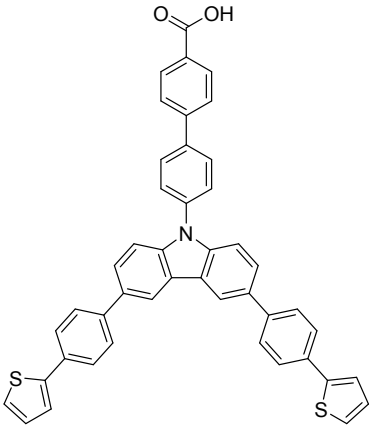
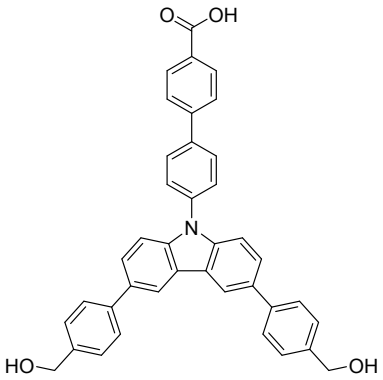
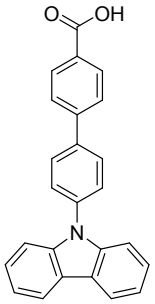
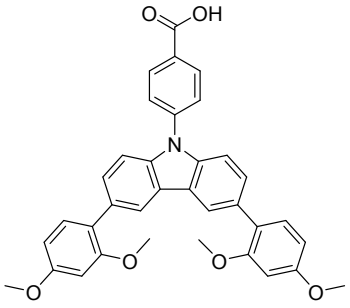
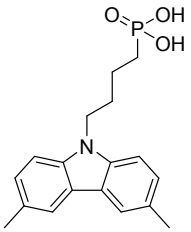
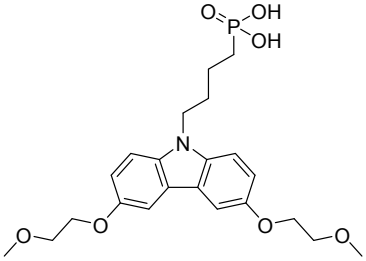


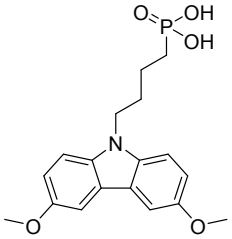
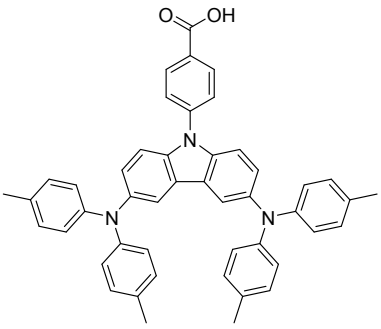
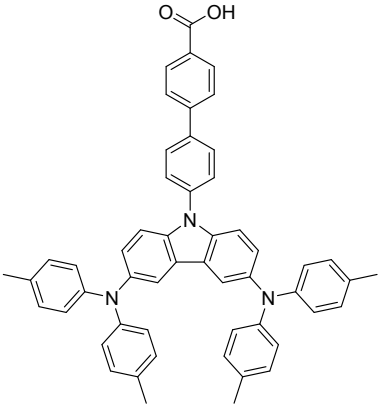
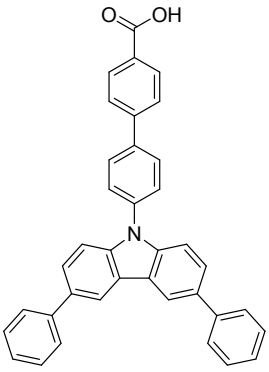
Figure S1. The data distribution of the processed dataset used for model training.

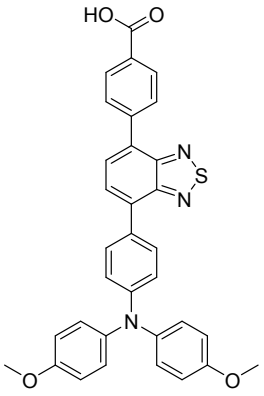
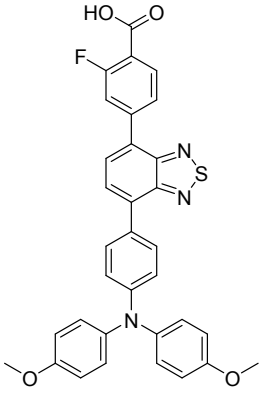
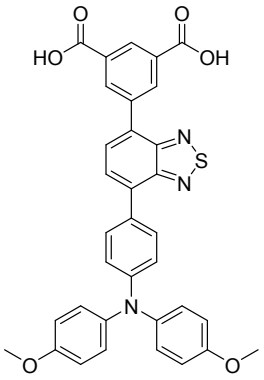
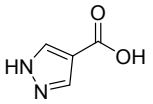
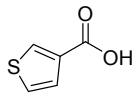
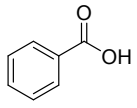
Table S2 Information of SAM molecular

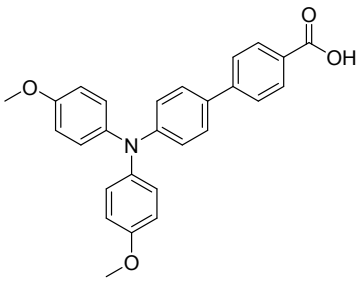
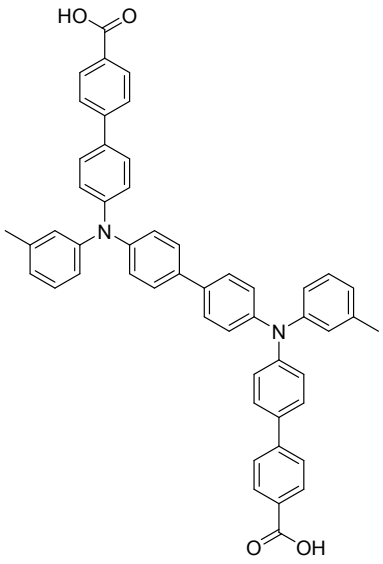
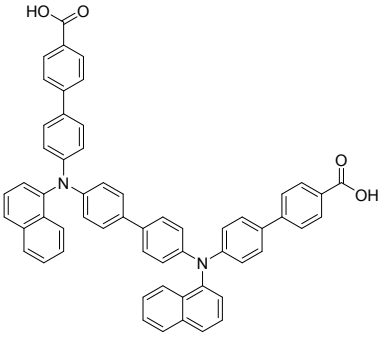
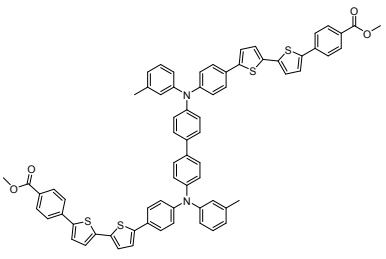
SAM Name	SMILES	Molecular Structure
2PACz	OP(CCN1C2=C(C=CC=C2)C3=C1C=CC=C3)(O)=O	
9CPA	OC(CCN1C2=C(C=CC=C2)C3=C1C=CC=C3)=O	
9CAA	OC(CN1C2=C(C=CC=C2)C3=C1C=CC=C3)=O	
BCB-C4PA	OP(CCCCN1C2=C(C(C=CC=C3)=C3C=C2)C4=C1C=CC=C4)(O)=O	
EA-46	OC(C(C=C1)=CC=C1C(S2)=CC(CCCCCC)=C2C(S3)=CC(CCCCCC)=C3C4=CC(OC)=C(OC)C(OC)=C4)=O	
EA-49	OC(C(C=C1)=CC=C1C(S2)=CC(CCCCCC)=C2C(S3)=CC(CCCCCC)=C3C4=CC(OC)=CC(OC)=C4)=O	
EA-41	BrC1=C(CCCCCC)C=C(C2=C(CCCCCC)C=C(C3=CC=C(C(O)=O)C=C3)S2)S1	

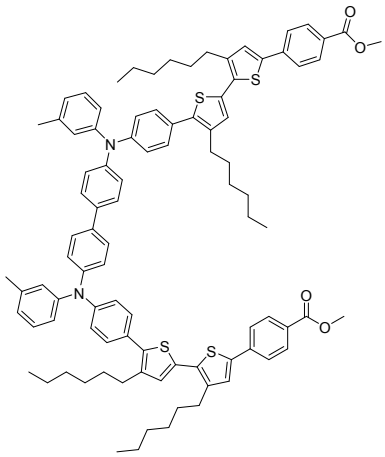
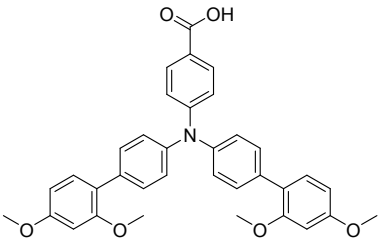
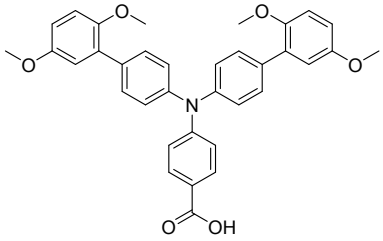
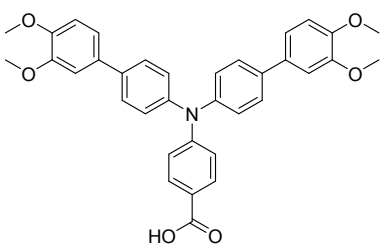
MeO-2PACz	OP(CCN1C2=C(C=C(OC)C=C2)C3=C1C=CC(OC)=C3)(O)=O	
CBSA	ClC1=CC=C(S(=O)(O)=O)C=C1	
HC-A1	CCC(CCCC)COC(C=C1)=CC=C1C(C=C2)=CC3=C2N(C4=CC=C(C(O)=O)C=C4)C5=C3C=C(C6=CC=C(OCC(CCCC)CC)C=C6)C=C5	
HC-A4	CC1(C)C2=C(C=CC=C2)C(C1=C3)=CC=C3N(C4=CC(C(C=CC=C5)=C5C6(C)C)=C6C=C4)C7=C8C(C=CC=C8)=C(C(O)=O)C=C7	
SAM4	O=C(O)C(C=C1)=CC=C1C(C=C2)=CC=C2N3C4=C(C=C(C5=CC=CS5)C=C4)C6=C3C=CC(C7=CC=CS7)=C6	
SAM5	O=C(O)C(C=C1)=CC=C1C(C=C2)=CC=C2N3C4=C(C=C(C5=CC=C(C6=CC=CS6)C=C5)C=C4)C7=C3C=CC(C8=CC=C(C9=CC=CS9)C=C8)=C7	

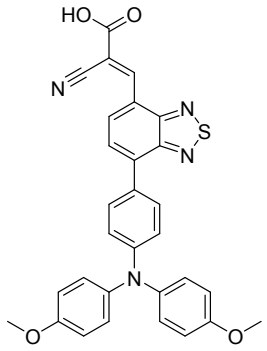
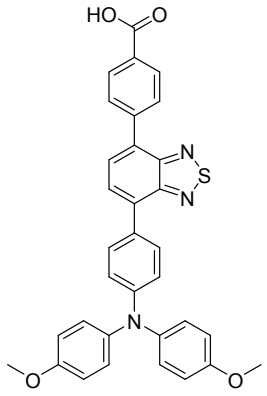
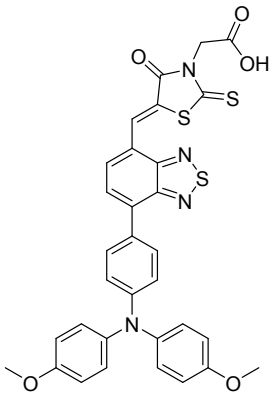
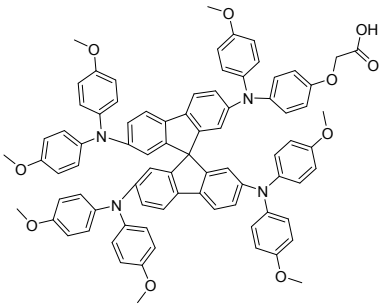
SAM6	O=C(O)C(C=C1)=CC=C1C(C=C2)=CC=C2N3C4=C(C=C(C5=CC=C(CO)C=C5)C=C4)C6=C3C=CC(C7=CC=C(CO)C=C7)=C6	
SAM7	O=C(O)C(C=C1)=CC=C1C(C=C2)=CC=C2N3C4=C(C=CC=C4)C5=C3C=CC=C5	
EADR03	OC(C(C=C1)=CC=C1N2C3=C(C=C(C4=C(OC)C=C(OC)C=C4)C=C3)C5=C2C=CC(C6=C(OC)C=C(OC)C=C6)=C5)=O	
Me-4PACz	OP(CCCCN1C2=CC=C(C)C=C2C3=CC(C)=CC=C31)(O)=O	
GM-4PACz	OP(CCCCN1C2=C(C=C(OCCOC)C=C2)C3=C1C=CC(OCCOC)=C3)(O)=O	

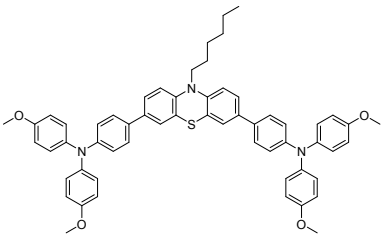
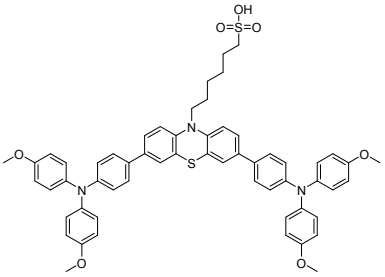
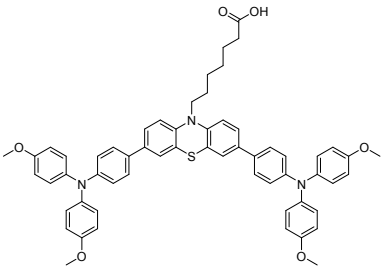
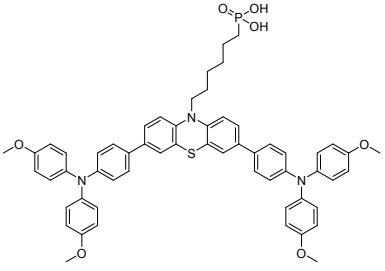
MeO-4PACz	OP(CCCCN1C2=CC=C(C=C2C3=CC(OC)=CC=C31)OC)(O)=O	
SAM1	O=C(O)C(C=C1)=CC=C1N2C3=C(C=C(N(C4=CC=C(C)C=C4)C5=CC=C(C)C=C5)C=C3)C6=C2C=CC(N(C7=CC=C(C)C=C7)C8=CC=C(C)C=C8)=C6	
SAM2	CC(C=C1)=CC=C1N(C2=CC=C(C)C=C2)C(C=C3)=CC4=C3N(C5=CC=C(C6=CC=C(C(O)=O)C=C6)C=C5)C7=C4C=C(N(C8=CC=C(C)C=C8)C9=CC=C(C)C=C9)C=C7	
SAM3	O=C(O)C(C=C1)=CC=C1C(C=C2)=CC=C2N3C4=C(C=C(C5=CC=CC=C5)C=C4)C6=C3C=CC(C7=CC=CC=C7)=C6	

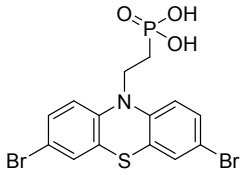
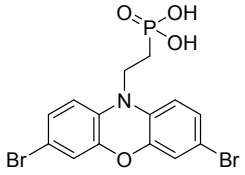
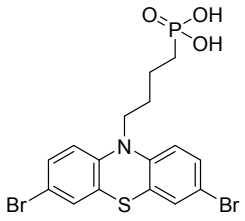
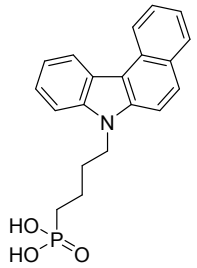
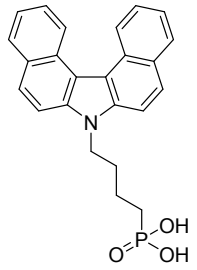
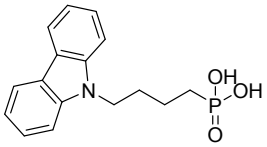
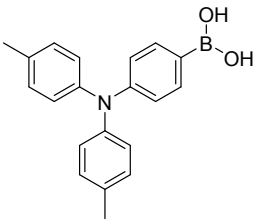
TBT-BA	COC(C=C1)=CC=C1N(C2=CC=C(OC)C=C2)C(C=C3)=CC=C3C4=CC=C(C5=CC=C(C(O)=O)C=C5)C6=NSN=C64	
TBT-FBA	COC(C=C1)=CC=C1N(C2=CC=C(OC)C=C2)C(C=C3)=CC=C3C4=CC=C(C5=CC(F)=C(C(O)=O)C=C5)C6=NSN=C64	
TBT-DBA	COC(C=C1)=CC=C1N(C2=CC=C(OC)C=C2)C(C=C3)=CC=C3C4=CC=C(C5=CC(C(O)=O)=CC(C(O)=O)=C5)C6=NSN=C64	
4-PCA	O=C(O)C1=CNN=C1	
3-TA	O=C(O)C1=CSC=C1	
BA	OC(C1=CC=CC=C1)=O	

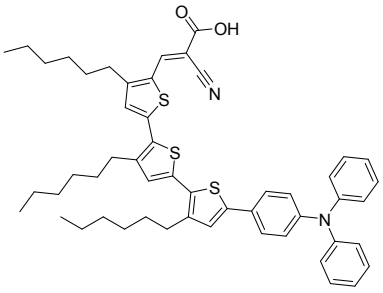
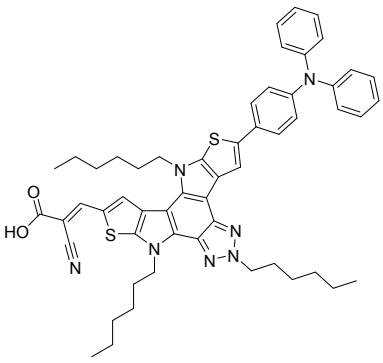
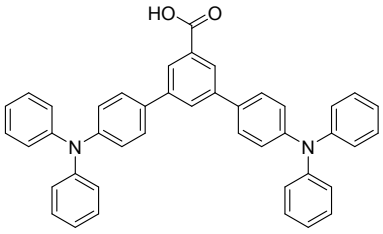
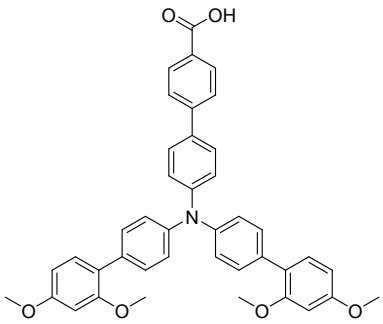
MC-45	O=C(C(C=C1)=CC=C1C2=CC=C(N(C3=CC=C(OC)C=C3)C4=CC=C(OC)C=C4)C=C2)O	
MC-54	CC1=CC=CC(N(C2=CC=C(C3=CC=C(C(O)=O)C=C3)C=C2)C4=CC=C(C5=CC=C(N(C6=CC=C(C7=CC=C(C(O)=O)C=C7)C=C6)C8=CC=CC(C)=C8)C=C5)C=C4)=C1	
MC-55	O=C(C(C=C1)=CC=C1C(C=C2)=CC=C2N(C3=C(C=CC=C4)C4=CC=C3)C5=CC=C(C6=CC=C(N(C7=CC=C(C8=CC=C(C(O)=O)C=C8)C=C7)C9=CC=CC%10=C9C=CC=C%10)C=C6)C=C5)O	
EA-59	CC1=CC=CC(N(C2=CC=C(C3=CC=C(N(C4=CC=C(C5=CC=C(C6=CC=C(C7=CC=C(C(OC)=O)C=C7)S6)S5)C=C4)C8=CC=CC(C)=C8)C=C3)C=C2)C9=CC=C(C%10=CC=C(C%11=CC=C(C%12=CC=C(C(OC)=O)C=C%12)S%11)S	

	%10)C=C9)=C1	
EA-60	CC1=CC=CC(N(C2=CC=C(C3=CC=C(N(C4=CC=C(C5=C(CCCCCC)C=C(C6=C(CCCCCC)C=C(C7=CC=C(C(OC)=O)C=C7)S6)S5)C=C4)C8=CC=CC(C)=C8)C=C3)C=C2)C9=CC=C(C%10=C(CCCCCC)C=C(C%11=C(CCCCCC)C=C(C%12=CC=C(C(OC)=O)C=C%12)S%11)S%10)C=C9)=C1	
RC-24	O=C(O)C1=CC=C(N(C2=CC=C(C3=C(OC)C=C(OC)C=C3)C=C2)C4=CC=C(C5=C(OC)C=C(OC)C=C5)C=C4)C=C1	
RC-25	O=C(O)C1=CC=C(N(C2=CC=C(C3=C(OC)C=CC(OC)=C3)C=C2)C4=CC=C(C5=C(OC)C=CC(OC)=C5)C=C4)C=C1	
RC-34	O=C(O)C1=CC=C(N(C2=CC=C(C3=CC=C(OC)C(OC)=C3)C=C2)C4=CC=C(C5=CC=C(OC)C(OC)=C5)C=C4)C=C	

MPA-BT-CA	COC(C=C1)=CC=C1N(C2=CC=C(OC)C=C2)C3=CC=C(C4=CC=C(/C=C(C(O)=O)\C#N)C5=NSN=C54)C=C3	
MPA-BT-BA	COC(C=C1)=CC=C1N(C2=CC=C(OC)C=C2)C3=CC=C(C4=CC=C(C5=CC=C(C(O)=O)C=C5)C6=NSN=C64)C=C3	
MPA-BT-RA	COC(C=C1)=CC=C1N(C2=CC=C(OC)C=C2)C3=CC=C(C4=CC=C(/C=C(C(N5CC(O)=O)=O)=O)\SC5=S)C6=NSN=C64)C=C3	
Spiro-Acid	OC(COC(C=C1)=CC=C1N(C2=CC=C(OC)C=C2)C3=CC4=C(C5=CC=C(N(C6=CC=C(OC)C=C6)C7=CC=C(OC)C=C7)C=C5C4(C8=CC(N(C9=CC=C(OC)C=C9)C%10=CC=C(OC)C=C%10)=CC=C8%11)C%12=C%11C=CC(N(C%13=CC=C(OC)C=C%13)C%14=CC=C(OC)C=C%14)=C%12)	

	C=C3)=O	
TPT-H6	COC1=CC=C(N(C2=CC=C(C3=CC=C(N4CCCCCC)C(SC5=C4C=CC(C6=CC=C(N(C7=CC=C(OC)C=C7)C8=CC=C(OC)C=C8)C=C6)=C5)=C3)C=C2)C9=CC=C(OC)C=C9)C=C1	
TPT-S6	COC1=CC=C(N(C2=CC=C(C3=CC=C(N4CCCCCCS(=O)(O)=O)C(SC5=C4C=CC(C6=CC=C(N(C7=CC=C(OC)C=C7)C8=CC=C(OC)C=C8)C=C6)=C5)=C3)C=C2)C9=CC=C(OC)C=C9)C=C1	
TPT-C6	COC(C=C1)=CC=C1N(C2=CC=C(C=C2)OC)C(C=C3)=CC=C3C(C=C4SC5=C6C=CC(C7=CC=C(C=C7)N(C8=CC=C(C=C8)OC)C9=CC=C(C=C9)OC)=C5)=CC=C4N6CCCCCCC(O)=O	
TPT-P6	COC1=CC=C(N(C2=CC=C(C3=CC=C(N4CCCCCCP(O)(O)=O)C(SC5=C4C=CC(C6=CC=C(N(C7=CC=C(OC)C=C7)C8=CC=C(OC)C=C8)C=C6)=C5)=C3)C=C2)C9=CC=C(	

	OC)C=C9)C=C1	
Br-2EPT	O=P(O)(O)CCN1C2=CC=C(Br)C=C2SC3=C1C=CC(Br)=C3	
Br-2EPO	O=P(O)(CCN1C2=CC=C(C=C2OC3=C1C=CC(Br)=C3)Br)O	
2BrPTZPA	BrC1=CC=C2N(C3=C(SC2=C1)C=C(Br)C=C3)CCCCP(O)(O)=O	
CbzPh	OP(CCCCN1C2=C(C(C=CC=C3)=C3C=C2)C4=C1C=CC=C4)(O)=O	
CbzNaph	OP(CCCCN1C2=C(C(C=CC=C3)=C3C=C2)C4=C1C=CC5=C4C=CC=C5)(O)=O	
4PACz	OP(CCCCN1C2=C(C=CC=C2)C3=C1C=CC=C3)(O)=O	
MTPA-BA	OB(C1=CC=C(C=C1)N(C2=CC=C(C=C2)C)C3=CC=C(C=C3)C)O	

FNE29	CCCCCCC1=C(C2=CC(CCCCCC)=C(C3=CC(CCCCCC)=C(/C=C(C(O)=O)\C#N)S3)S2)SC(C4=CC=C(N(C5=CC=CC=C5)C6=CC=CC=C6)C=C4)=C1	
DT-1	CCCCCCN1N=C2C(C(C(C=C(C3=CC=C(N(C4=CC=CC=C4)C5=CC=CC=C5)C=C3)S6)=C6N7CCCCCC)=C7C8=C2N(CCCCCC)C9=C8C=C(/C=C(C(O)=O)\C#N)S9)=N1	
TPA	O=C(O)C1=CC(C2=CC=C(N(C3=CC=CC=C3)C4=CC=CC=C4)C=C2)=CC(C5=CC=C(N(C6=CC=CC=C6)C7=CC=CC=C7)C=C5)=C1	
MC-43	OC(C(C=C1)=CC=C1C2=CC=C(N(C3=CC=C(C4=CC=C(OC)C=C4OC)C=C3)C5=CC=C(C6=C(OC)C=C(OC)C=C6)C=C5)C=C2)=O	

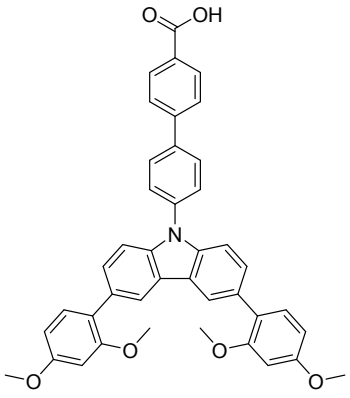
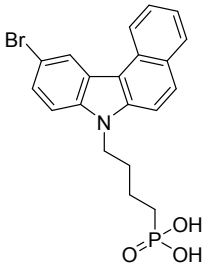
EADR04	COC(C=C1)=CC(OC)=C1C(C=C2)=CC3=C2N(C4=CC=C(C5=CC=C(C(O)=O)C=C5)C=C4)C6=C3C=C(C7=C(OC)C=C(OC)C=C7)C=C6	
BCBBR-C4PA	OP(CCCCN1C2=C(C(C=CC=C3)=C3C=C2)C4=C1C=CC(Br)=C4)(O)=O	

Table S3 The 26 features used to input ML models and their meanings

Input features	Description
Cs	ratio of Cs cations to all A-site cations in the perovskite layer
FA	ratio of FA cations to all A-site cations in the perovskite layer
MA	ratio of MA cations to all A-site cations in the perovskite layer
Pb	ratio of Pb cations to all B-site cations in the perovskite layer
Sn	ratio of Sn cations to all B-site cations in the perovskite layer
I	Ratio of I anions to all anions in the perovskite layer
Br	Ratio of Br anions to all anions in the perovskite layer
Cl	Ratio of CI anions to all anions in the perovskite layer
E	The HOMO level offset between the perovskite layer and the SAM
Concentration(mg/mL)	Concentration of the solvent
wash	Whether the SAM was washed（0: unwashed; 1: washed）
NiO_2	Whether the NiO_2 was used（0: no used; 1: used）
ethanol, toluene, isopropanol, tetrahydrofuran, 2-Methoxyethanol, chlorobenzene, CH_2Cl_2	Whether the solvent was used（0: no used; 1: used）
C_{60}, BCP, $PC_{60}BM$, PCBM, $PC_{61}BM$, PEAI, ALD-SnO_2	Whether the electron transport layer material was used（0: no used; 1: used）

Table S4 Hyperparameter tuning results of full dataset

Model	hyperparameter tuning result
RDKit&XGBoost	learning_rate: 0.93; n_estimators: 400; max_depth: 7; subsample: 0.9; colsample_bytree: 0.05; gamma: 0.57; min_child_weight: 5; reg_alpha: 0.1; reg_lambda: 1
256&XGBoost	learning_rate: 0.58; n_estimators: 400; max_depth: 7; subsample: 0.9; colsample_bytree: 0.3; gamma: 0.67; min_child_weight: 5; reg_alpha: 0.1; reg_lambda: 1
512&XGBoost	learning_rate: 0.58; n_estimators: 400; max_depth: 7; subsample: 0.9; colsample_bytree: 0.45; gamma: 0.5; min_child_weight: 5; reg_alpha: 0.1; reg_lambda: 1
1024&XGBoost	learning_rate: 0.22; n_estimators: 400; max_depth: 11; subsample: 0.9; colsample_bytree: 0.42; gamma: 0.35; min_child_weight: 5; reg_alpha: 0.1; reg_lambda: 1
RDKit&RF	n_estimators: 400; max_depth: 10; min_samples_split: 3; max_features: 0.6; random_state: 2
256&RF	n_estimators: 400; max_depth: 10; min_samples_split: 8; max_features: 0.85; random_state: 2
512&RF	n_estimators: 400; max_depth: 8; min_samples_split: 10; max_features: 0.8; random_state: 2
1024&RF	n_estimators: 400; max_depth: 11; min_samples_split: 2; max_features: 0.69; random_state: 2
RDKit&GBDT	n_estimators: 400; learning_rate: 0.68; max_depth: 3; min_samples_split: 2; min_samples_leaf: 1; max_features: None; subsample: 1; alpha: 0.9; loss: ‘squared_error’; random_state: 5
256&GBDT	n_estimators: 400; learning_rate: 0.45; max_depth: 4; min_samples_split: 2; min_samples_leaf: 1; max_features:

	None; subsample: 1; alpha: 0.9; loss: ‘squared_error’; random_state: 5
512&GBDT	n_estimators: 400; learning_rate: 0.58; max_depth: 4; min_samples_split: 2; min_samples_leaf: 1; max_features: None; subsample: 1; alpha: 0.9; loss: ‘squared_error’; random_state: 5
1024&GBDT	n_estimators: 400; learning_rate: 0.76; max_depth: 5; min_samples_split: 2; min_samples_leaf: 1; max_features: None; subsample: 0.7; alpha: 0.9; loss: ‘squared_error’; random_state: 5

Table S5 10-fold cross-validation performance metrics of 12 models for PCE prediction

		RMSE	R^2	r	MAE
XGBoost	RDKit	1.954 ± 0.592	0.6085 ± 0.1834	0.8375 ± 0.0941	1.493 ± 0.469
	Morgan_256	1.925 ± 0.606	0.5925 ± 0.2461	0.8158 ± 0.1379	1.510 ± 0.497
	Morgan_512	2.017 ± 0.821	0.5459 ± 0.3466	0.7765 ± 0.1991	1.579 ± 0.552
	Morgan_1024	1.695 ± 0.714	0.6649 ± 0.2680	0.8576 ± 0.1456	1.329 ± 0.531
RF	RDKit	2.134 ± 0.596	0.5277 ± 0.2328	0.7759 ± 0.1510	1.607 ± 0.441
	Morgan_256	2.112 ± 0.714	0.5317 ± 0.2611	0.7525 ± 0.1875	1.589 ± 0.496
	Morgan_512	2.147 ± 0.713	0.5148 ± 0.2697	0.7366 ± 0.1967	1.639 ± 0.483
	Morgan_1024	1.929 ± 0.650	0.6019 ± 0.2403	0.7974 ± 0.1623	1.436 ± 0.442
GBDT	RDKit	2.159 ± 0.688	0.5096 ± 0.2485	0.7799 ± 0.1026	1.649 ± 0.528
	Morgan_256	1.974 ± 0.403	0.5938 ± 0.1512	0.7994 ± 0.1003	1.476 ± 0.288
	Morgan_512	2.043 ± 0.644	0.5740 ± 0.1922	0.8088 ± 0.1034	1.586 ± 0.516
	Morgan_1024	2.569 ± 0.634	0.1984 ± 0.6469	0.7545 ± 0.1227	2.025 ± 0.496

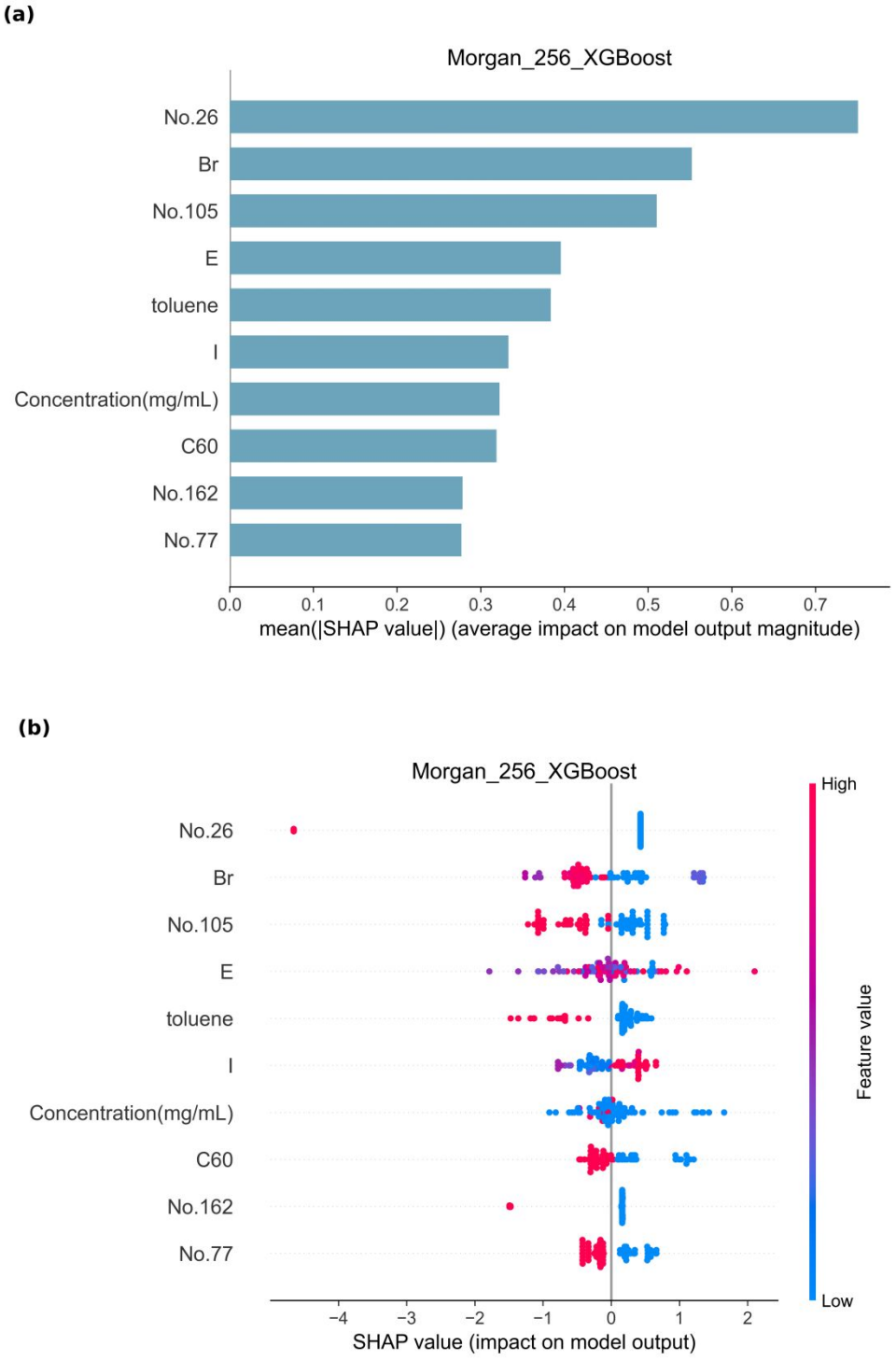


Figure S2. Importance and contribution of the top 10 features to Morgan_256&XGBoost model decision. a) Summary bar plot of Morgan_256&XGBoost . b) Swarm summary plot of Morgan_256&XGBoost

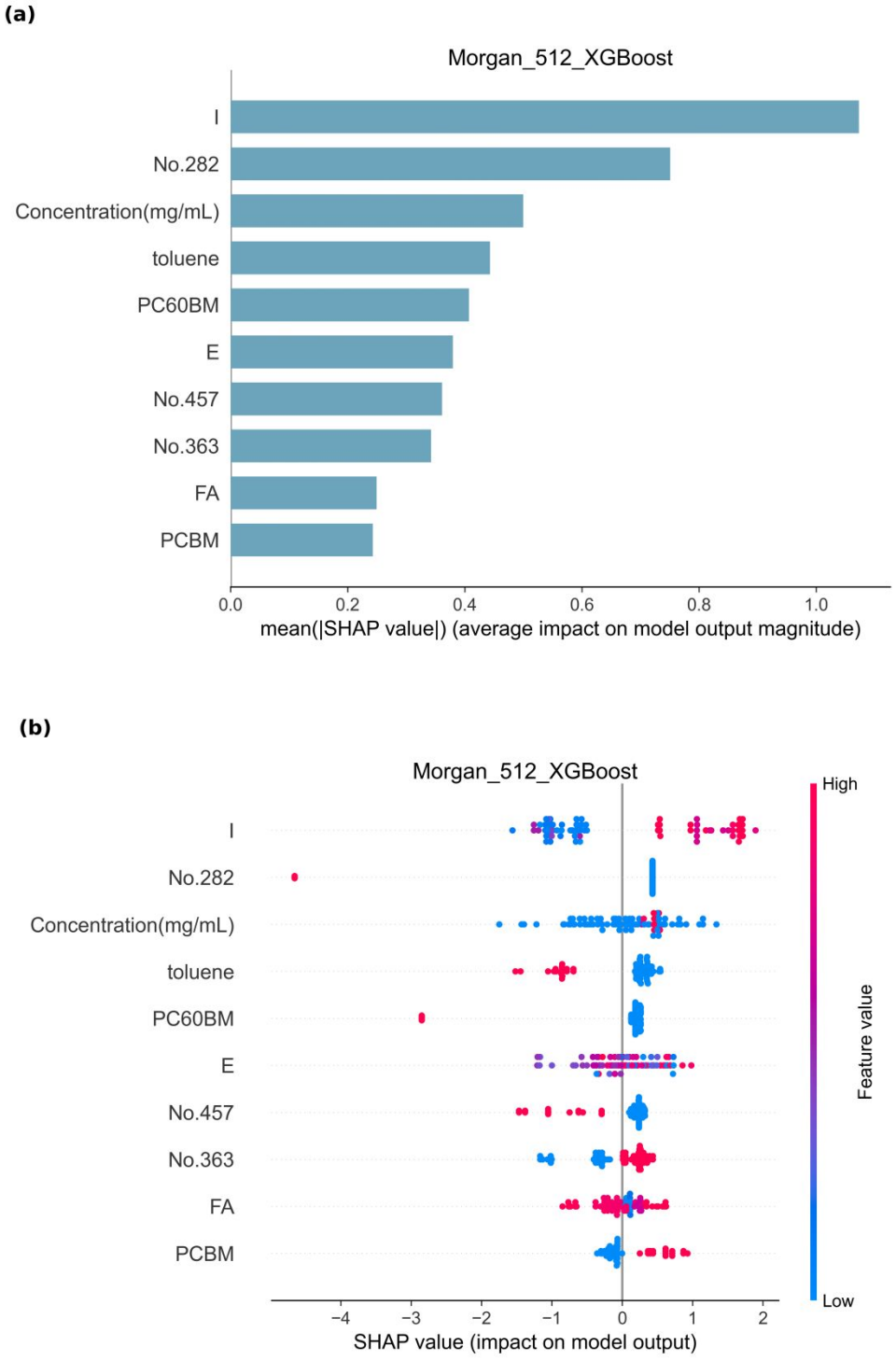


Figure S3. Importance and contribution of the top 10 features to Morgan_512&XGBoost model decision. a) Summary bar plot of Morgan_512&XGBoost . b) Swarm summary plot of Morgan_512&XGBoost

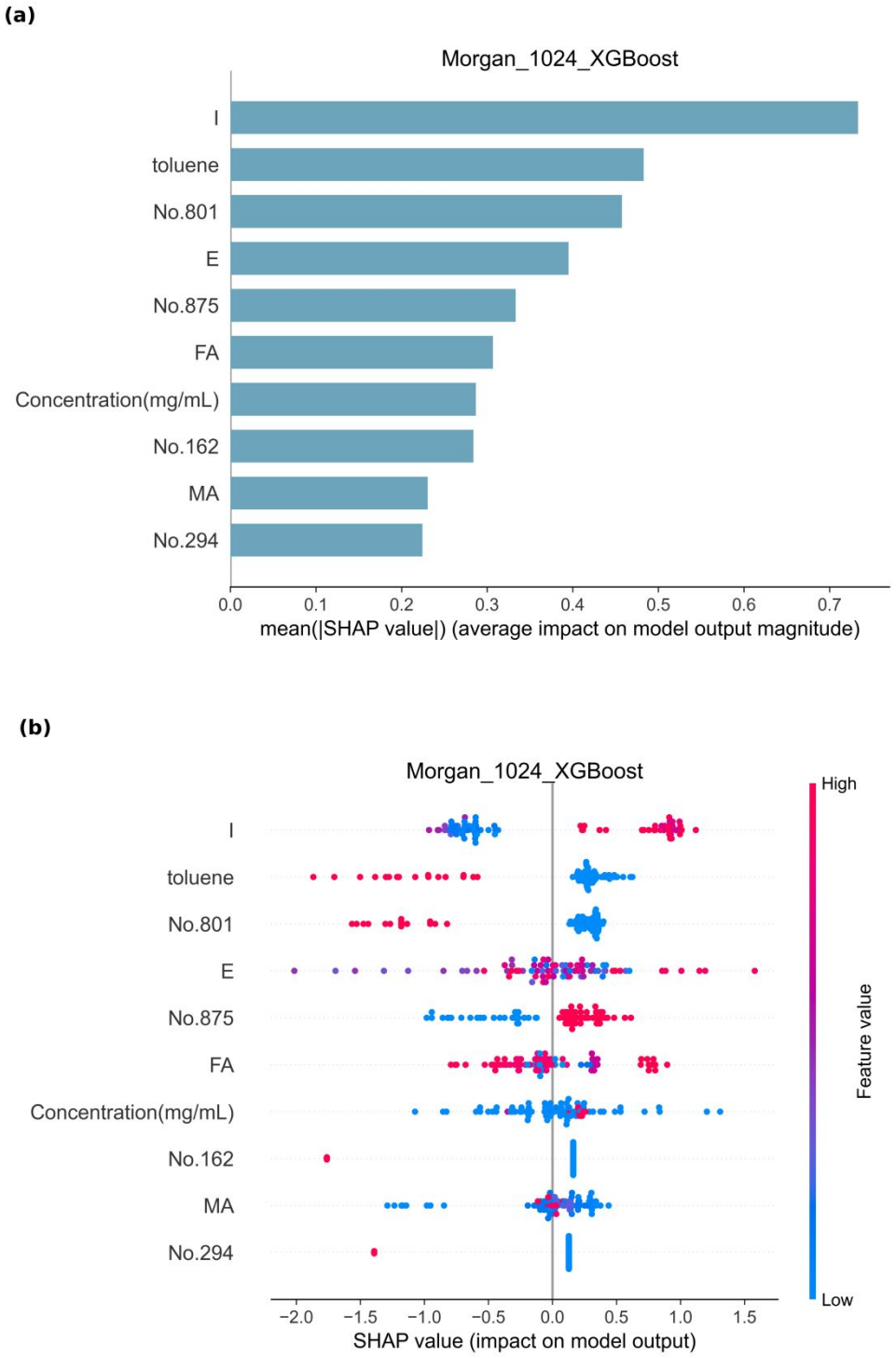


Figure S4. Importance and contribution of the top 10 features to Morgan_1024&XGBoost model decision. a) Summary bar plot of Morgan_1024&XGBoost . b) Swarm summary plot of Morgan_1024&XGBoost

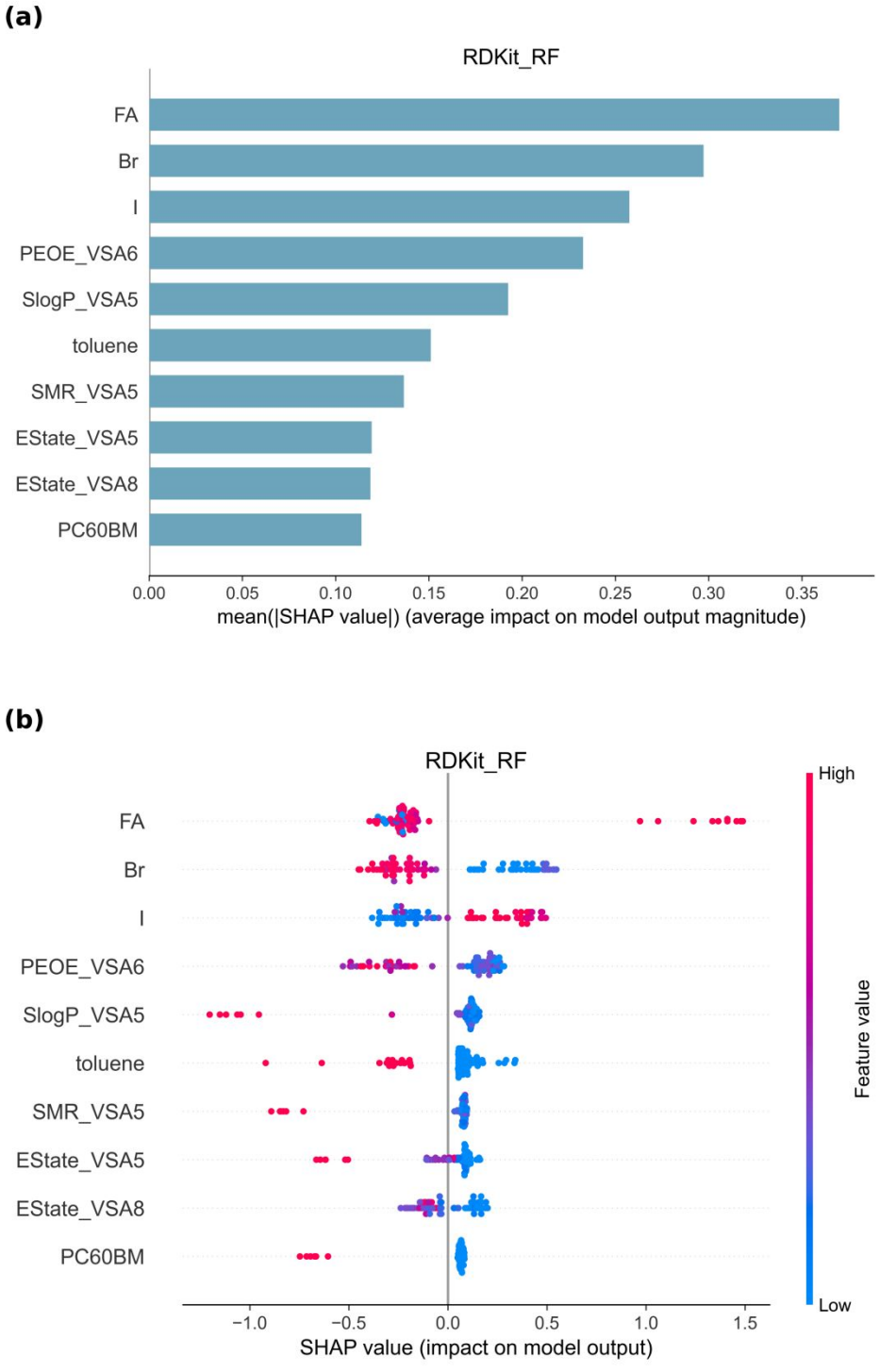


Figure S5. Importance and contribution of the top 10 features to RDKit&RF model decision. a) Summary bar plot of RDKit&RF . b) Swarm summary plot of RDKit&RF

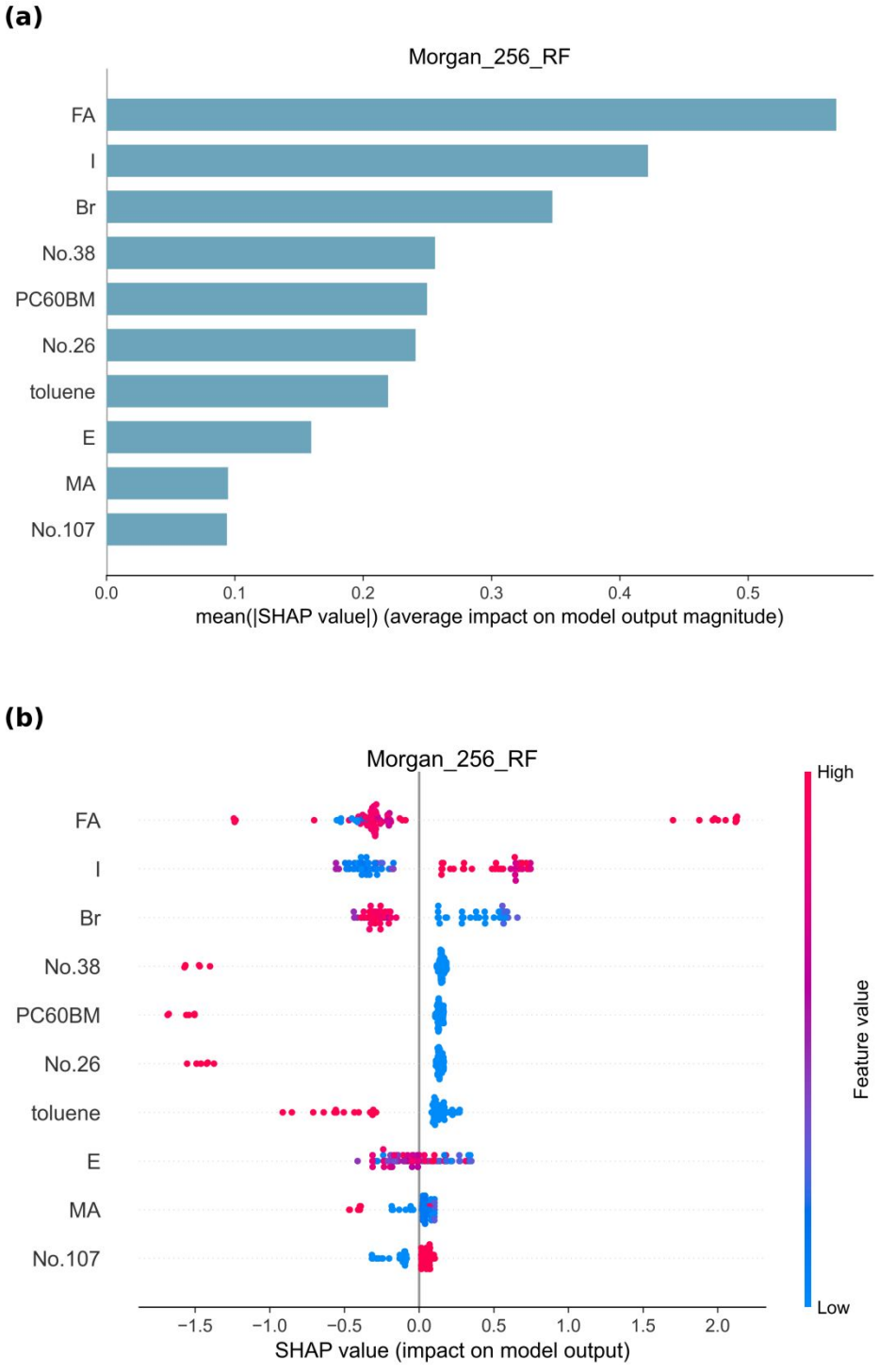


Figure S6. Importance and contribution of the top 10 features to Morgan_256&RF model decision. a) Summary bar plot of Morgan_256&RF . b) Swarm summary plot of Morgan_256&RF

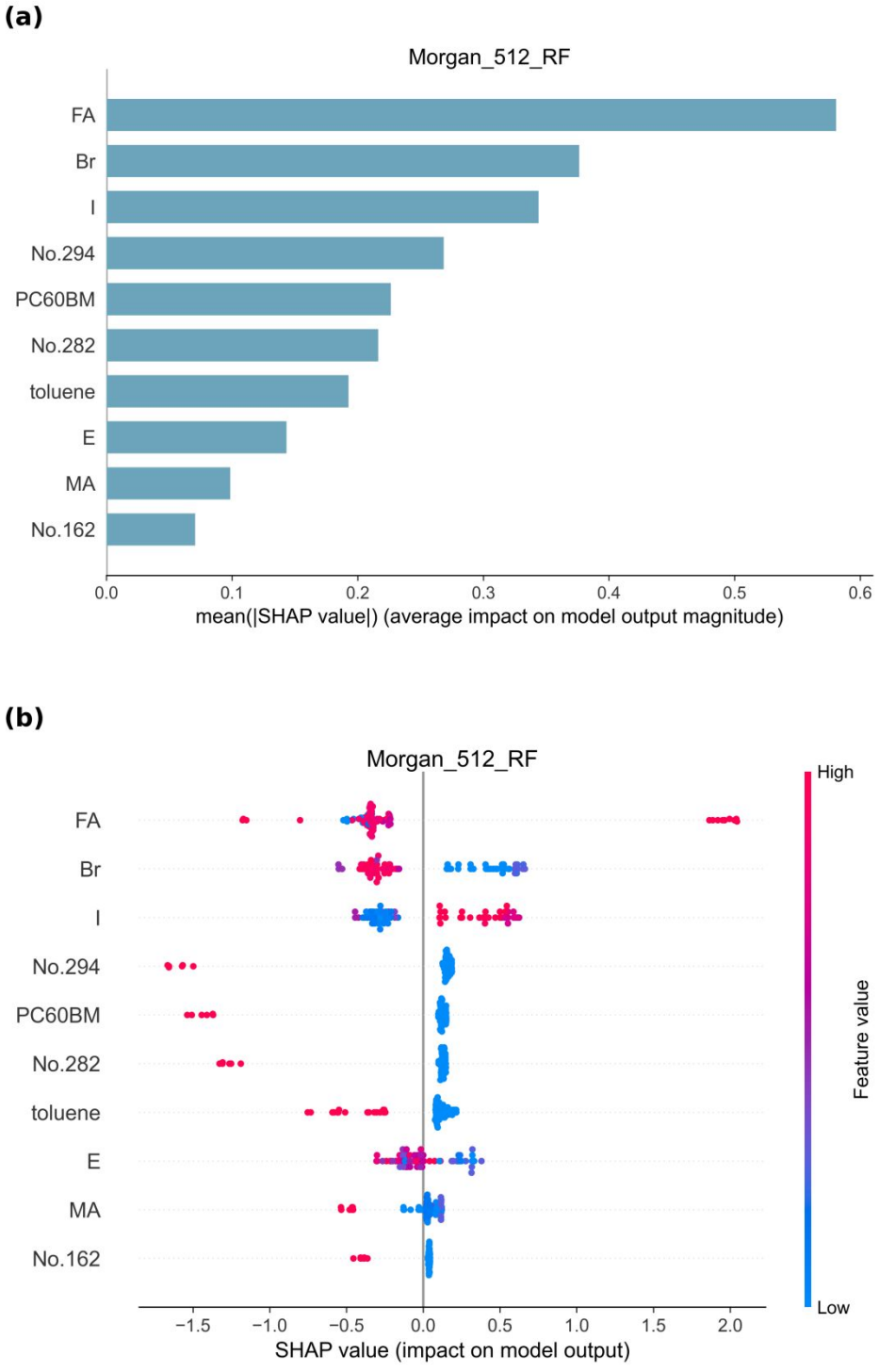


Figure S7. Importance and contribution of the top 10 features to Morgan_512&RF model decision. a) Summary bar plot of Morgan_512&RF . b) Swarm summary plot of Morgan_512&RF

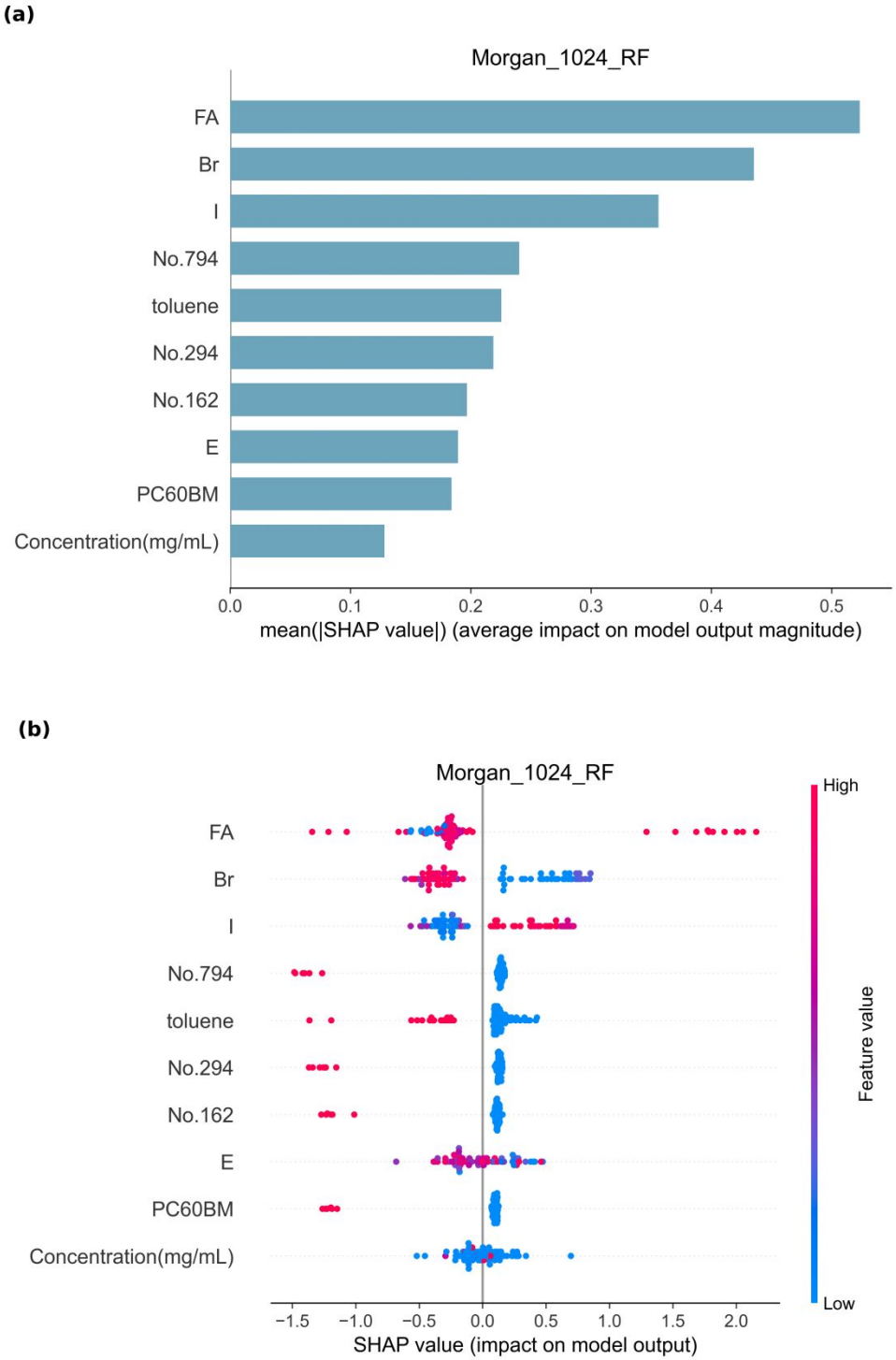


Figure S8. Importance and contribution of the top 10 features to Morgan_1024&RF model decision. a) Summary bar plot of Morgan_1024&RF . b) Swarm summary plot of Morgan_1024&RF

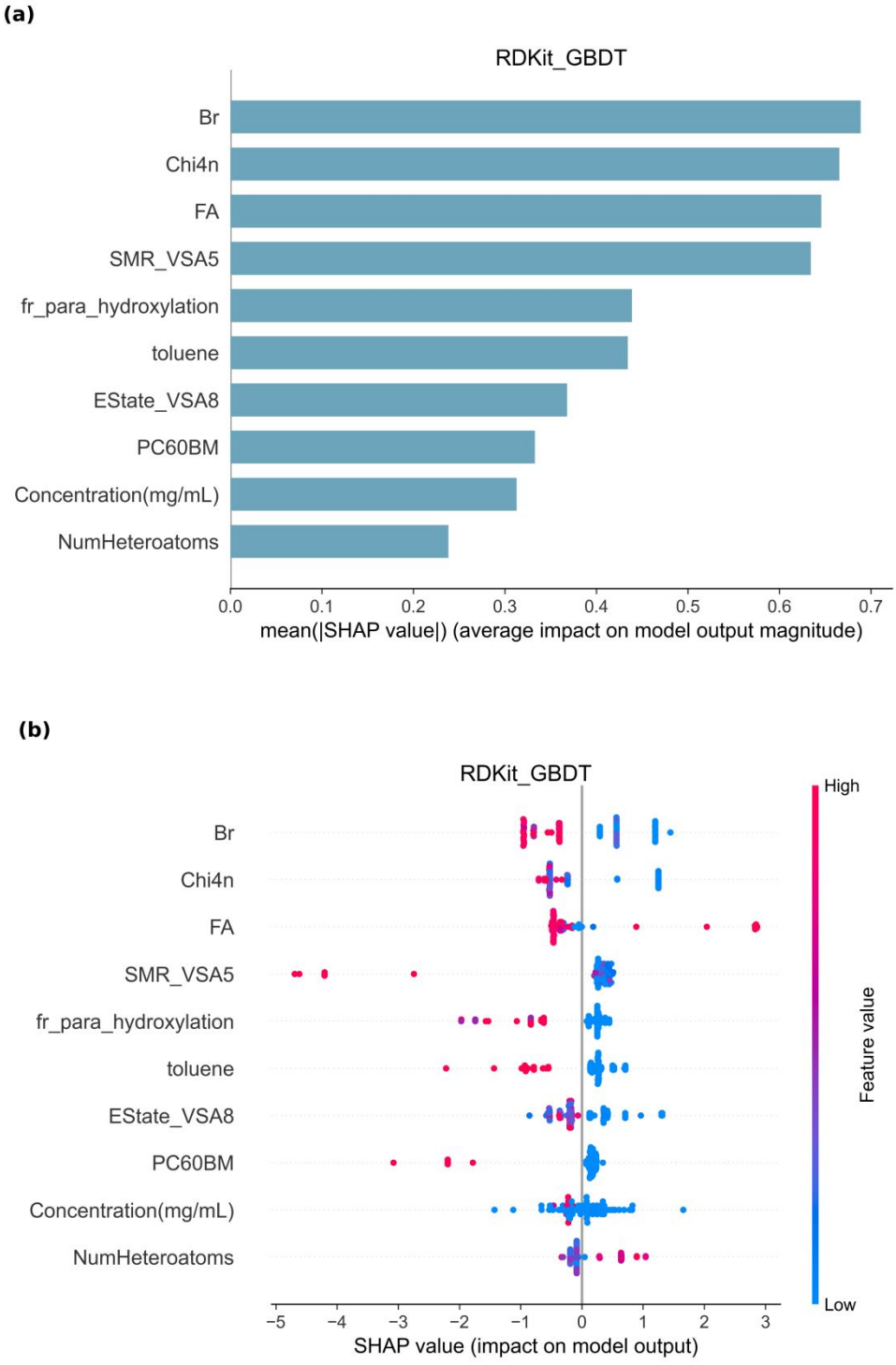


Figure S9. Importance and contribution of the top 10 features to RDKit&GBDT model decision. a) Summary bar plot of RDKit&GBDT . b) Swarm summary plot of RDKit&GBDT

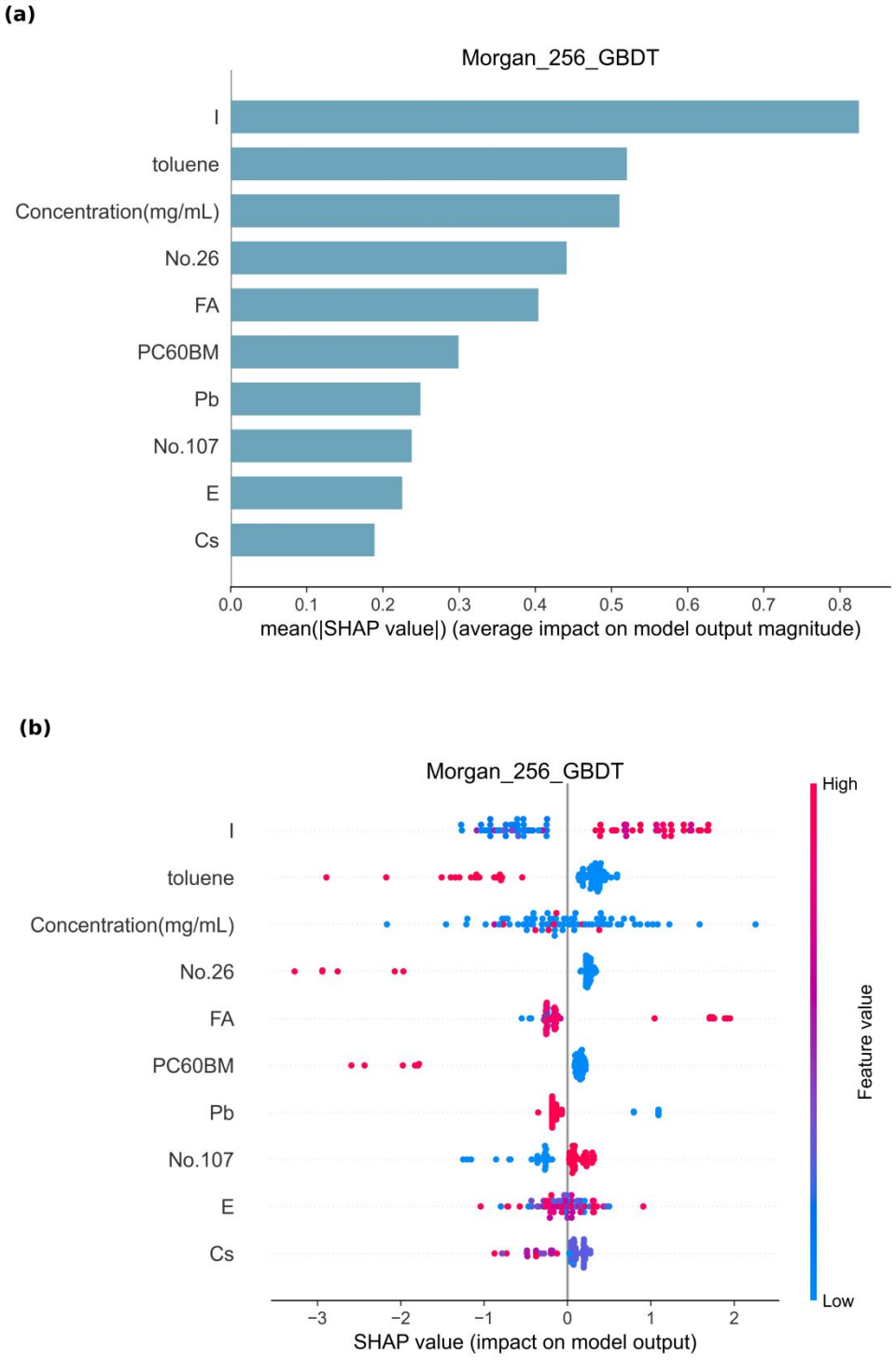


Figure S10. Importance and contribution of the top 10 features to Morgan_256&GBDT model decision. a) Summary bar plot of Morgan_256&GBDT . b) Swarm summary plot of Morgan_256&GBDT

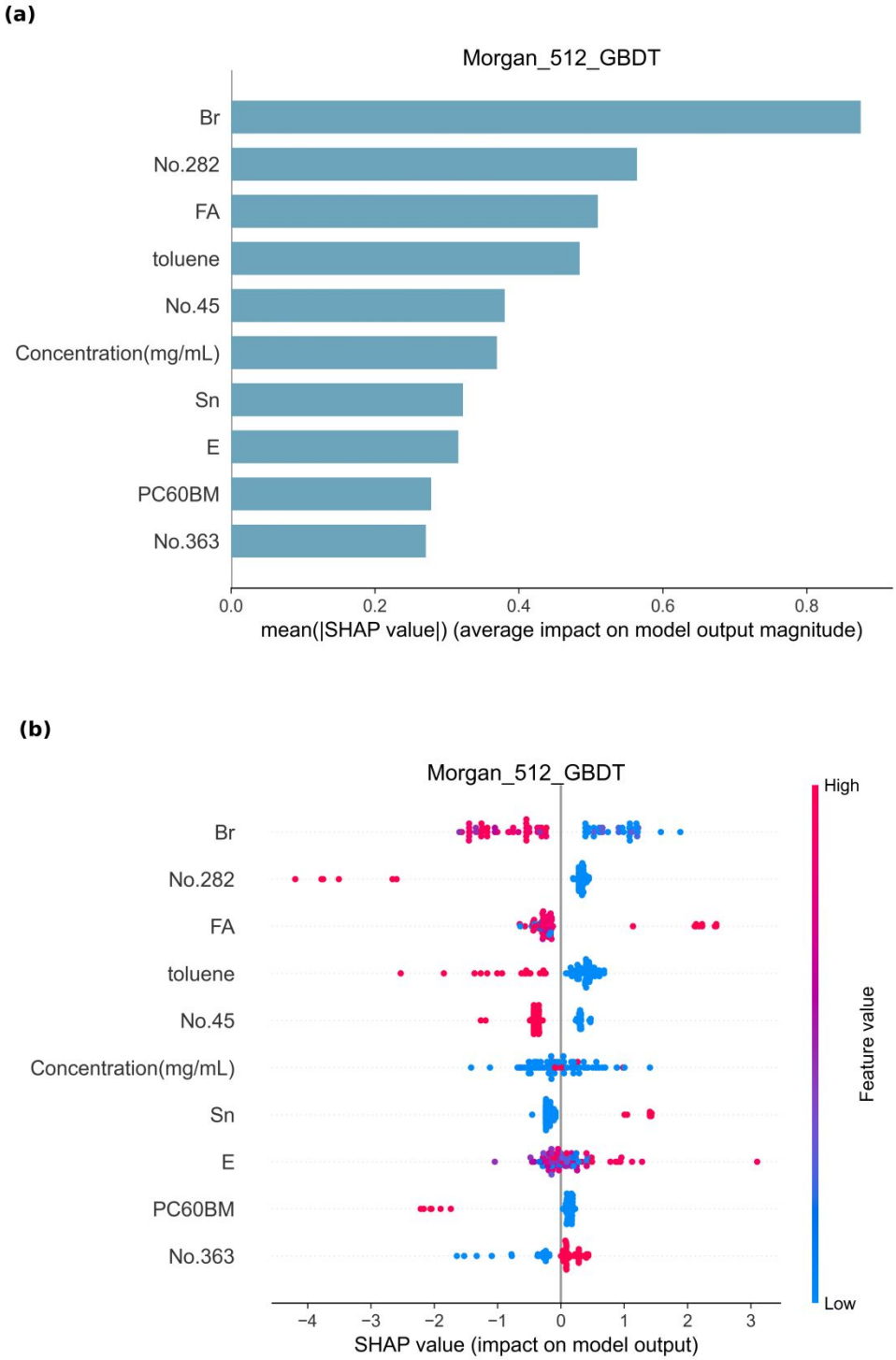


Figure S11. Importance and contribution of the top 10 features to Morgan_512&GBDT model decision. a) Summary bar plot of Morgan_512&GBDT . b) Swarm summary plot of Morgan_512&GBDT

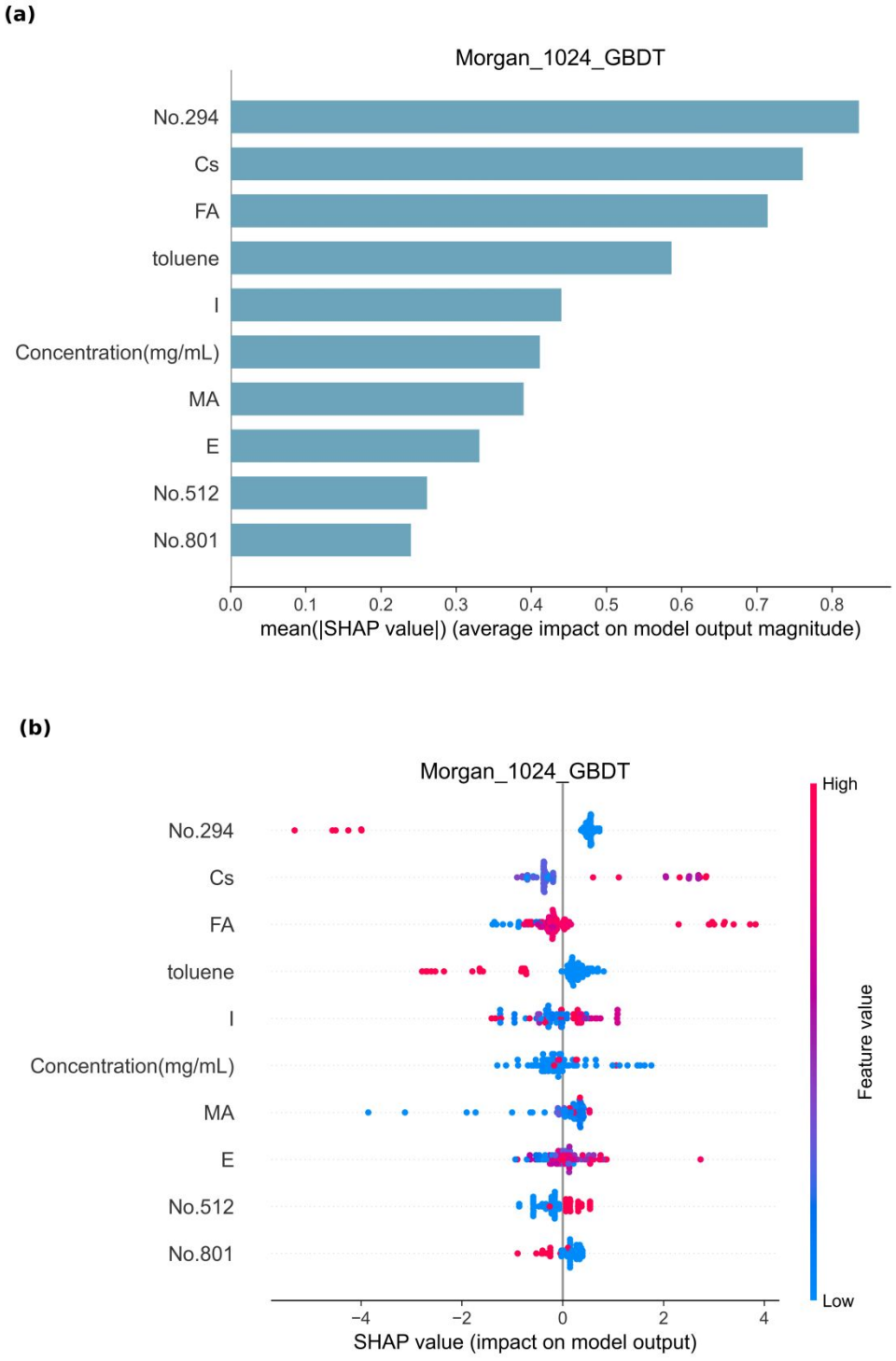


Figure S12. Importance and contribution of the top 10 features to Morgan_1024&GBDT model decision. a) Summary bar plot of Morgan_1024&GBDT . b) Swarm summary plot of Morgan_1024&GBDT

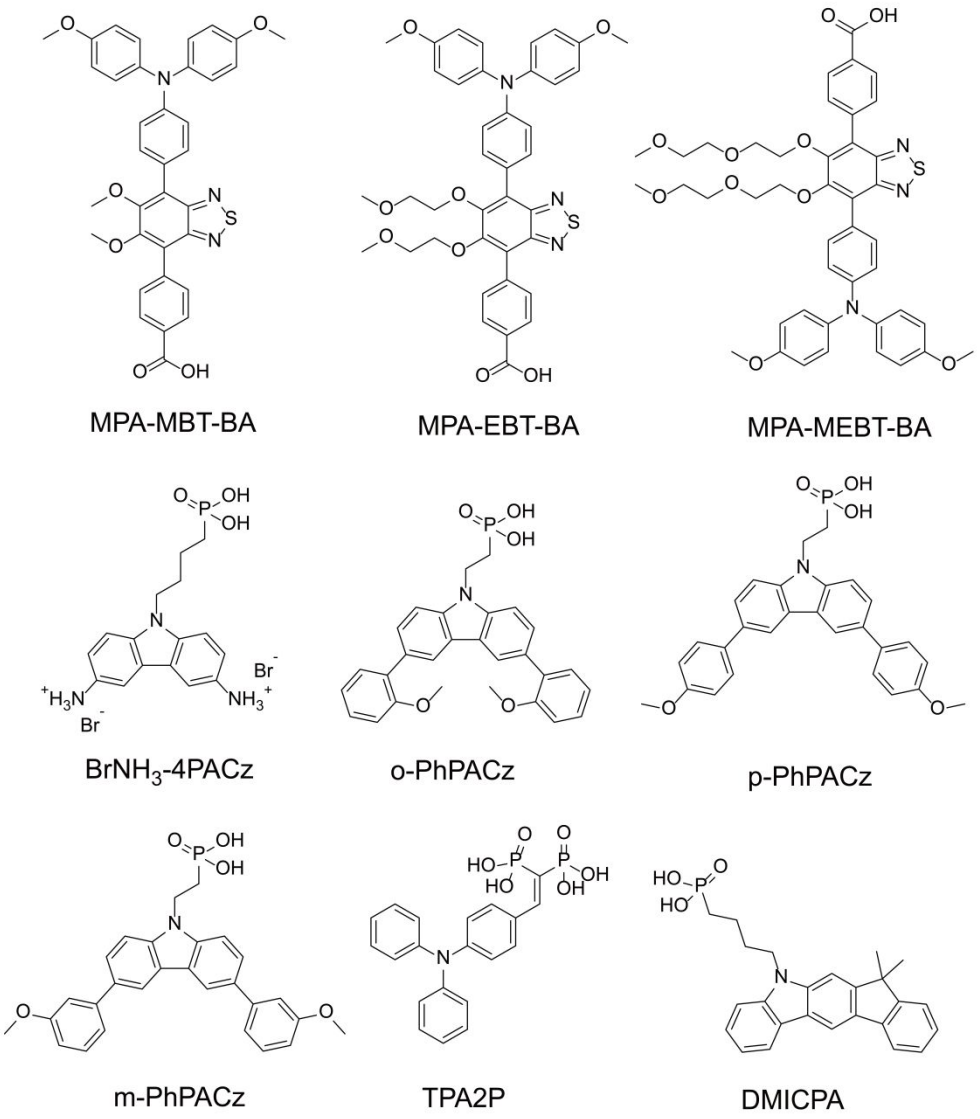


Figure S13. SAM molecular structures for model validation

REFERENCES

[1] A. Abid, P. Rajamanickam, and E. Diau, “Design of a simple bifunctional system as a self-assembled monolayer (SAM) for inverted tin-based perovskite solar cells,” Chemical Engineering Journal, vol. 477, Dec. 2023.

[2] W. Wang, K. Wei, L. Yang, J. Deng, J. Zhang, and W. Tang, “Dynamic self-assembly of small molecules enables the spontaneous fabrication of hole conductors at perovskite/electrode interfaces for over 22% stable inverted perovskite solar cells,” Materials Horizons, vol. 10, no. 7, pp. 2609 – 2617, Jul. 2023.

[3] E. Arkan et al., “Effect of functional groups of self assembled monolayer molecules on the performance of inverted perovskite solar cell,” Materials Chemistry and Physics, vol. 254, Nov. 2020.

[4] Z. Li et al., “Simple and robust phenoxazine phosphonic acid molecules as self-assembled hole selective contacts for high-performance inverted perovskite solar cells,” Nanoscale, vol. 15, no. 4, pp. 1676 – 1686, Jan. 2023.

[5] Z. Zhang et al., “Anchoring Charge Selective Self-Assembled Monolayers for Tin-Lead Perovskite Solar Cells,” Advanced Materials, vol. 36, no. 18, May 2024.

[6] A. Al-Ashouri et al., “Conformal monolayer contacts with lossless interfaces for perovskite single junction and monolithic tandem solar cells,” Energy & Environmental Science, vol. 12, no. 11, pp. 3356 – 3369, Nov. 2019.

[7] W. Wang, K. Wei, L. Yang, J. Deng, J. Zhang, and W. Tang, “Dynamic self-assembly of small molecules enables the spontaneous fabrication of hole conductors at perovskite/electrode interfaces for over 22% stable inverted perovskite solar cells,” Materials Horizons, vol. 10, no. 7, pp. 2609 – 2617, July 2023.

[8] E. Arkan et al., “Effect of functional groups of self assembled monolayer

molecules on the performance of inverted perovskite solar cell," Materials Chemistry and Physics, vol. 254, Nov. 2020.

[9] Y. Guo et al., "Efficient inverted perovskite solar cells with a low-temperature processed NiOx/SAM hole transport layer," Journal of Materials Chemistry C, vol. 12, no. 4, pp. 1507 - 1515, Jan. 2024.

[10] J. Zhang et al., "Elimination of Interfacial Lattice Mismatch and Detrimental Reaction by Self-Assembled Layer Dual-Passivation for Efficient and Stable Inverted Perovskite Solar Cells," Advanced Energy Materials, vol. 12, no. 18, May 2022.

[11] N. Phung et al., "Enhanced Self-Assembled Monolayer Surface Coverage by ALD NiO in p-i-n Perovskite Solar Cells," ACS Applied Materials & Interfaces, vol. 14, no. 1, pp. 2166 - 2176, Jan. 2022.

[12] G. Kim et al., "Enhancing Surface Modification and Carrier Extraction in Inverted Perovskite Solar Cells via Self-Assembled Monolayers," Nano Materials, vol. 14, no. 2, Jan. 2024.

[13] D. González et al., "Exploring the Interactions at the Interface: Tailoring Carbazole-Based Self-Assembled Molecules with Varying Functional Groups for Enhancing the Performance of Inverted Perovskite Solar Cells," Solar RRL, vol. 8, no. 12, June 2024.

[14] X. Luo, X. Liu, J. Nakazaki, H. Segawa, Y. Wang, and L. Han, "Facile Posttreatment of Self-Assembled Monolayers for Efficient Inverted Perovskite Solar Cells," Solar RRL, vol. 8, no. 3, Feb. 2024.

[15] H. Zhou et al., "Glycol Monomethyl Ether-Substituted Carbazolyl Hole-Transporting Material for Stable Inverted Perovskite Solar Cells with Efficiency of 25.52 %," Angewandte Chemie-International Edition, vol. 63, no. 33, Aug. 2024.

[16] T. Li et al., "In Situ Dehydration Condensation of Self-Assembled Molecules Enables Stabilization of CsPbI3 Perovskites for Efficient Photovoltaics," Advanced Functional Materials, vol. 34, no. 51, Dec. 2024.

[17] D. González et al., “Influence of the carbazole moiety in self-assembling molecules as selective contacts in perovskite solar cells: interfacial charge transfer kinetics and solar-to-energy efficiency effects,” Nanoscale Advances, vol. 5, no. 23, pp. 6542 – 6547, Nov. 2023.

[18] Y. Zhou et al., “Interfacial Modification of NiOx for Highly Efficient and Stable Inverted Perovskite Solar Cells,” Advanced Energy Materials, vol. 14, no. 25, July 2024.

[19] X. Yuan et al., “Modulation on electrostatic potential to build a firm bridge at NiOx/ perovskite interface for efficient and stable perovskite solar cells,” Journal of Energy Chemistry, vol. 96, pp. 249 – 258, Sept. 2024.

[20] E. Yalcin et al., “Monodentate versus Bidentate Anchoring Groups in Self-Assembling Molecules (SAMs) for Robust p-i-n Perovskite Solar Cells,” ACS Applied Materials & Interfaces, vol. 15, no. 49, pp. 57153 – 57164, Dec. 2023.

[21] E. Arkan et al., “Performance enhancement of inverted perovskite solar cells through interface engineering by TPD based bidentate self-assembled monolayers,” Optical Materials, vol. 105, July 2020.

[22] E. Aktas et al., “Role of Terminal Group Position in Triphenylamine-Based Self-Assembled Hole-Selective Molecules in Perovskite Solar Cells,” ACS Applied Materials & Interfaces, vol. 14, no. 15, pp. 17461 – 17469, Apr. 2022.

[23] Q. Liao et al., “Self-assembled donor-acceptor hole contacts for inverted perovskite solar cells with an efficiency approaching 22%: The impact of anchoring groups,” Journal of Energy Chemistry, vol. 68, pp. 87 – 95, May 2022.

[24] W. Li, M. Cariello, M. Méndez, G. Cooke, and E. Palomares, “Self-Assembled Molecules for Hole-Selective Electrodes in Highly Stable and Efficient Inverted Perovskite Solar Cells with Ultralow Energy Loss,” ACS Applied Energy Materials, vol. 6, no. 3, pp. 1239 – 1247, Jan. 2023.

[25] E. Li et al., “Bonding Strength Regulates Anchoring-Based Self-Assembly

Monolayers for Efficient and Stable Perovskite Solar Cells," Advanced Functional Materials, vol. 31, no. 35, Aug. 2021.

[26] A. Ullah et al., "Novel Phenothiazine-Based Self-Assembled Monolayer as a Hole Selective Contact for Highly Efficient and Stable p-i-n Perovskite Solar Cells," Advanced Energy Materials, vol. 12, no. 2, Jan. 2022.

[27] A. Ullah et al., "Versatile Hole Selective Molecules Containing a Series of Heteroatoms as Self-Assembled Monolayers for Efficient p-i-n Perovskite and Organic Solar Cells," Advanced Functional Materials, vol. 32, no. 49, Dec. 2022.

[28] Z. Li et al., "Simple and robust phenoxazine phosphonic acid molecules as self-assembled hole selective contacts for high-performance inverted perovskite solar cells," Nanoscale, vol. 15, no. 4, pp. 1676 - 1686, Jan. 2023.

[29] W. Jiang, F. Li, M. Li, F. Qi, F. Lin, and A. Jen, "π-Expanded Carbazoles as Hole-Selective Self-Assembled Monolayers for High-Performance Perovskite Solar Cells," Angewandte Chemie-International Edition, vol. 61, no. 51, Dec. 2022.

[30] H. Guo et al., "Neglected acidity pitfall: boric acid-anchoring hole-selective contact for perovskite solar cells," National Science Review, vol. 10, no. 5, Apr. 2023.

[31] H. Cheng and Z. Huang, "Self-assembled organic molecules with a fused aromatic ring as hole-transport layers for inverted perovskite solar cells: the effect of linkers on performance," New Journal of Chemistry, vol. 48, no. 15, pp. 6833 - 6841, Apr. 2024.

[32] E. Yalcin et al., "Semiconductor self-assembled monolayers as selective contacts for efficient PiN perovskite solar cells," Energy & Environmental Science, vol. 12, no. 1, pp. 230 - 237, Jan. 2019.

[33] E. Aktas et al., "Understanding the perovskite/self-assembled selective contact interface for ultra-stable and highly efficient p-i-n perovskite solar cells," Energy & Environmental Science, vol. 14, no. 7, pp. 3976 - 3985, July 2021.

[34] W. Wang et al., “Versatile Self-Assembled Molecule Enables High-Efficiency Wide-Bandgap Perovskite Solar Cells and Organic Solar Cells,” Advanced Energy Materials, vol. 13, no. 23, June 2023.